

Review

Stem-like cells at the center of CD4 T cell differentiation

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CD4 T cells orchestrate immune responses through differentiation into specialized helper subsets. Traditionally, CD4 T cell differentiation is described as a linear process in which naïve CD4 T cells commit to distinct effector lineages upon activation. However, growing evidence challenging this paradigm has provoked considerable debate about CD4 T cell plasticity. This review describes the emerging concept of stem-like CD4 T cells and how they update our understanding of the fundamental mechanisms that regulate CD4 T cell differentiation. We discuss how stem-like CD4 T cells play a crucial role as precursor cells to distinct effector subsets and explore their implications in cancer, infections, and autoimmunity. Finally, we address how stem-like CD4 T cells might resolve longstanding questions about CD4 T cell plasticity, and propose alternative differentiation models that incorporate this population in chronic diseases.

CD4 T cell subsets control the balance between protective immunity and disease

CD4 T cell differentiation is a central event in adaptive immune responses. Upon antigen recognition, naïve CD4 T cells integrate cytokine signals produced by innate immune cells to undertake specialized effector programs [1]. Each CD4 T cell subset performs specific functions to either promote cellular immunity or support high-affinity antibody production [1–3]. This differentiation process allows CD4 T cells to coordinate innate and adaptive immune cells and dynamically tailor the best response to a wide variety of pathogens and immunological challenges. However, while they are crucial for host protection, inappropriate or dysregulated CD4 T cell differentiation can induce pathological conditions, including asthma, diabetes, autoimmunity, and organ rejection [4–7]. Given this central role in controlling the balance between protective immunity and disease, understanding the mechanisms that regulate CD4 T cell differentiation fate remains a fundamental question in immunology.

The paradigm of CD4 T cell differentiation and specialized helper subsets stems from the seminal observations of Mossman and Coffman who first described two CD4 T cell clones with distinct cytokine profiles: interferon γ (IFN- γ)-producing type 1 T helper (Th1) cells and interleukin-4 (IL-4)-producing Th2 cells [8]. This finding was a major conceptual advance in the field and introduced the concept of CD4 T cell heterogeneity and specific lineage functions. Since this initial discovery, subsequent studies have established that Th1 cells promote innate immune cell phagocytosis, improve CD8 T cell activation and cytotoxicity, and enhance the magnitude and longevity of CD8 T cell memory [9–11]. By contrast, Th2 cells play an essential role in controlling immunity to helminths and extracellular pathogens, primarily by stimulating the function of eosinophils and granulocytes at the site of infection [12]. Our understanding of this initial model has evolved with the discovery of additional CD4 T cell subsets. In 1995, Sakaguchi and colleagues identified a population of CD25⁺ regulatory CD4 T cells that lacked classical Th1 or Th2 cytokine production and instead suppressed self-reactivity and maintained immune tolerance to host tissues [13–15]. More recently, Th17 cells, defined by their IL-17 production,

Highlights

CD4 T cell differentiation is crucial for balancing effective immune defense and disease pathology.

Stem-like CD4 T cells are present among antigen-specific CD4 T cells. This unique population retains self-renewal and differentiation capacity, and helps to sustain effector responses that mediate disease outcomes across chronic conditions.

Stem-like CD4 T cells may also exhibit multipotency to differentiate into various helper subsets, and thus provide a potential explanation for CD4 T cell plasticity.

Targeting stem-like CD4 T cell differentiation could represent a novel therapeutic strategy for immune modulation across various diseases.

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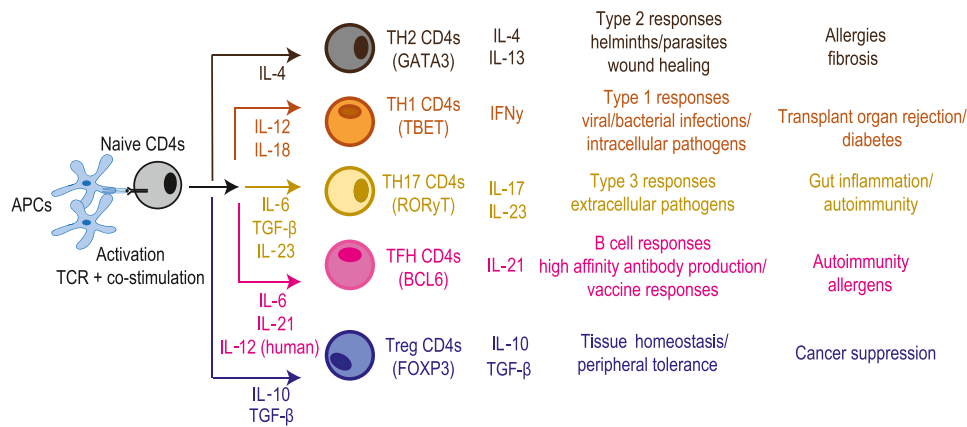


were identified as key regulators of extracellular bacteria and gut microbiota [16,17]. Finally, T follicular helper (T_{fh}) cells emerged as being crucial for supporting productive B cell responses, germinal center reactions, and high-affinity antibody production [18–20]. Beyond these core CD4 T cell subsets, other CD4 T cell lineages with unique cytokine profiles and functions, including Th9, T follicular regulatory (T_{fr}), and Th22 cells, have been identified in specific immunological contexts, leading to an increasingly complex view of CD4 T cell differentiation [21–24]. Further research will be necessary to refine our understanding of how these populations integrate with the broader CD4 T cell landscape. Together, these findings established the foundation for our current understanding of CD4 T cell heterogeneity, where naïve cells differentiate into specialized helper programs to promote specific immune responses. However, while they are essential for host protection, effector CD4 T cells can also contribute to immune pathology. Accumulating evidence indicates that Th1 and Th17 cells are implicated in organ rejection, diabetes, gut immunopathology, and tissue inflammation, while Th2 and T_{fh} cells contribute to asthma and allergic diseases [4,6,7,25,26]. Furthermore, regulatory T cells (Tregs) are known to suppress conventional CD4 and effector CD8 T cell responses, and thus reduce the efficacy of antitumor immunity and response to therapies [27,28]. Collectively, these findings highlight how CD4 T cell fate decisions control the balance between protective responses, tolerance, and autoimmunity. Understanding how CD4 T cells commit, maintain, or alter their differentiation fate is essential for developing novel strategies that target or redirect their effector functions. In this review we examine CD4 T cell differentiation in the context of chronic antigen stimulation, with a particular focus on stem-like CD4 T cells. We highlight the key functional properties of stem-like CD4 T cells, propose alternative models of differentiation, and explore their potential as therapeutic targets in chronic diseases.

CD4 T cell differentiation: fixed effector lineages or plastic states?

Given the central role of CD4 T cell differentiation in determining the outcome of immune responses, a central question in immunology remains whether committed CD4 T cell subsets can be reprogrammed into alternative lineages. Early models proposed that, once a naïve CD4 T cell adopted a specific effector fate, it remained fixed within that differentiation trajectory [29]. This concept was heavily supported by the expression of master transcription factors that define effector CD4 T cell subsets [30]. These transcription factors are not only necessary and sufficient to drive the commitment of naïve CD4 T cells to a particular fate but also actively repress alternative differentiation programs (Figure 1). For example, TBET, the Th1 master regulator, represses Th2 lineage commitment by a direct tyrosine-kinase interaction with GATA3 [31]. Conversely, GATA3 inhibits the Th1 program by modulating chromatin accessibility around IFN γ and other Th1-associated loci [32,33]. Similar transcriptional activator and repressor mechanisms stabilize the Th17 (ROR γ T), T_{fh} (BCL6), and Treg (FOXP3) lineages [34–37], reinforcing the model of terminally differentiated and mutually exclusive CD4 T cell subsets.

Despite the strict transcriptional regulation, accumulating evidence suggests that there is some degree of plasticity within committed CD4 T cell subsets, indicating the possibility to undertake alternative differentiation programs [38]. Indeed, transcription factor reporters and lineage-tracing experiments have shown that activated Tregs can lose FOXP3 expression and gain features of Th1 and Th17 effector cells, including transcription factor expression and cytokine production [35,39–42]. This level of plasticity and reprogramming capacity has also been reported in other helper lineages. Th17 cells can take on general features of Th1 cells under inflammatory conditions, and thus contribute to immunopathology [43,44]. Similarly, T_{fh} cells have been found to generate Th1 effector cells and take on Th1, Th2, or Th17 functional characteristics [45,46], while they also can gain FOXP3 expression and function as regulatory cells [47]. Although these



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Figure 1. CD4 T cell subsets control the balance between protective immunity and disease. Upon activation with cognate antigen presented on major histocompatibility complex II (MHC-II) by antigen-presenting cells (APCs), naïve CD4 T cells undergo effector differentiation into a specialized helper or regulatory subset depending on innate-derived cytokine signals. CD4 T cell subset differentiation depends on the expression of master transcription factors that both promote specific effector functions and repress alternative differentiation fates. These differentiated CD4 T cell subsets can enhance host protection or promote disease pathology through their effector functions across diverse immune responses. Abbreviations: IL, interleukin; TCR, T cell receptor; TFH, T follicular helper cell; TH1, TH2, TH17, type 1/2/17 T helper cells; Treg, regulatory T cell.

observations are apparent across many disease models, many studies support the framework that CD4 T cell lineages remain stable and resistant to full reprogramming into alternative fates [34,35,48–50]. These conflicting observations have led to multiple proposed models of CD4 T cell differentiation to reconcile the balance between lineage commitment and plasticity [3,51].

Discovery of CD4 T cells with stem-like capacity in several diseases

Naïve CD4 T cells commit to specific effector subsets early after activation, within the first rounds of division [51–53]. Whether this differentiation is terminal depends on the conceptual framework and remains an ongoing debate. Recent studies in cancer, transplant, asthma, and autoimmunity, spanning both human and mouse models, have identified a stem-like precursor CD4 T cell population that challenges this early commitment paradigm and more importantly may help to provide a unifying model for CD4 T cells that accounts for lineage plasticity [7,54–60]. These stem-like CD4 T cells are both activated and antigen-specific but preserve an uncommitted phenotypic and transcriptional state. Across disease settings, stem-like CD4 T cells consistently express high levels of T cell factor 1 (TCF1) and minimal or undetectable expression of master transcription factors. Phenotypically, stem-like CD4 T cells are also distinct from naïve cells. Despite retaining expression of TCF1, they upregulate an array of activation markers, chemokine receptors, transcription factors, and inhibitory receptors that are otherwise absent in resting naïve CD4 T cells. In mice, stem-like CD4 T cells are characterized by high expression of CD44, programmed cell death protein 1 (PD-1), Ly108, thymocyte selection-associated high mobility group box protein (TOX), CD73, inhibitor of DNA binding 3 (ID3), and G protein-coupled receptor 183 (GPR183), while in human patients PD-1, CD26, IL-7 receptor, TOX, and CD28 were crucial to define this population. They can also be distinguished from terminally differentiated effector subsets by the lack of expression of inhibitory receptors, effector molecules, and transcription factors associated with committed helper lineages, including CXCR6, CD39, CTLA-4, CXCR5, PSGL1, SLAMF1, B lymphocyte-induced maturation protein 1 (BLIMP-1), and CD38. While the expression of some of these markers varied across disease settings, likely reflecting differences in the tissue microenvironments, their core transcriptional identity was conserved [7,54–59].

Functionally, these cells retained remarkable proliferative and differentiation capacity to give rise to terminally differentiated effector CD4 T cell subsets, despite constant stimulation, making them indispensable to sustain antigen-specific CD4 T cell responses in chronic disease [7,54–59]. In fact, the loss of TCF1 or failure to establish this precursor population resulted in an early burst of effector CD4 T cells that failed to be maintained over time [55,56]. A series of key experiments demonstrated that stem-like CD4 T cells are the primary source of effector cells in each disease. It is worth noting that in each of these conditions the differentiation of stem-like CD4 T cells into terminal effector cells dictated the outcome of the response, and either improved protective immunity or contributed to disease pathology. For instance, in a transplantation model, transfer of a small number of antigen specific TCF1⁺ stem-like CD4 T cells was sufficient to generate a robust Th1 response that ultimately led to graft rejection [56]. By contrast, antigen-specific Th1 cells from the same donor failed to expand or further differentiate upon transfer, resulting in graft tolerance [56]. Similarly, in the experimental autoimmune encephalomyelitis (EAE) model, precursor CD4 T cells dramatically expanded and gave rise to effector Th17 cells that exacerbated immune pathology and gut inflammation [60]. Similar experiments have been reported in allergic and asthma models, where stem-like CD4 T cells uniquely led to the generation of effector Th2 CD4 T cells that exacerbated immunopathology, whereas terminally differentiated cells remained functionally limited [58].

Notably, this ability to preserve an uncommitted phenotype, long-term persistence and serve as precursors for more terminally differentiated populations is shared with stem-like CD8 T cells [60–66]. Furthermore, both stem-like CD4 and CD8 T cells also share preferential localization within secondary lymphoid tissues and the capacity to adapt to their environment [54,56,60–62,64,67]. While stem-like CD8 T cells predominantly give rise to cytotoxic effector and ultimately exhausted subsets, they are capable of differentiating into more functional CD8 T cell states given the right conditions [68–71]. As we discuss below, evidence now suggests that stem-like CD4 T cells also retain multipotency, and have the ability to differentiate into other helper lineages depending on the environment. Building on this concept, recent experiments using a persistent strain of lymphocytic choriomeningitis virus (LCMV) identified a TCF1⁺ precursor virus-specific CD4 T cell that, upon retransfer into infection matched mice, differentiated into both Th1 and Tfh CD4 T cells [55]. These findings demonstrate their ability to undertake multiple differentiation fates and suggest that this population may be the source of the observed plasticity across CD4 T cell subsets. Together, these studies defined a division of labor among antigen-specific CD4 T cells during chronic stimulation. Stem-like CD4 T cells sustain the long-term response through their proliferative and differentiation potential, while their terminally differentiated progeny lose proliferative capacity but acquire effector functions that ultimately determine immune outcomes.

Stem-like CD4 T cells retain differentiation potential beyond their priming environment

The experiments described above showed that stem-like CD4 T cells serve as precursors to the specialized effector subsets that are characteristic of each disease. However, whether these cells could reprogram into alternative differentiation fates was not tested. Recently, this hypothesis was examined with stem-like CD4 T cells in mouse tumor models and patients with kidney, prostate, or bladder cancer [54]. Among antigen experienced cells, stem-like CD4 T cells were the predominant population in draining lymph nodes and tumors. While these cells retained an activated but uncommitted phenotype, they demonstrated a remarkable capacity to proliferate and undertake various canonical CD4 T cell differentiation programs under polarizing conditions, even though some of these helper lineages were absent from the tumor microenvironment. This was in striking contrast to their effector counterparts from matched patient tumors that exhibited minimal proliferative potential and terminally differentiated states [54]. These findings suggest that stem-

like CD4 T cells retain the capacity to differentiate into various canonical helper subsets beyond their priming environment, despite being present within patient tumors for months to years. Similar observations were made across mouse tumor models, highlighting their functionality and differentiation flexibility. Using a system where subcutaneous tumors expressed the LCMV glycoprotein (GP), chronically stimulated GP-specific stem-like CD4 T cells gave rise to inducible FOXP3⁺ Tregs when transferred back into matched tumor-bearing mice. However, when transferred and challenged with LCMV infection, stem-like CD4 T cells generated robust Th1 and Tfh cells at frequencies comparable to the endogenous virus-specific CD4 T cells. By contrast, antigen-specific effector Tregs from tumor bearing mice retained their FOXP3⁺ differentiated state in both the tumor and following LCMV rechallenge. Similarly, stem-like lineage-negative CD4 T cells in oral tolerance models that generally give rise to inducible Tregs were shown to undertake alternative differentiation programs when placed in an inflammatory environment [7,72]. These data indicate that chronically stimulated stem-like CD4 T cells retain multipotency and can give rise to effector populations that are not characteristic of their priming environment. Further research will be necessary to determine whether this is a conserved feature across disease settings. Nonetheless, these findings strongly suggest that CD4 T cell plasticity is encapsulated within the stem-like population. The reported fate switching across CD4 T cell subsets may not reflect reprogramming of terminally differentiated effectors but rather the inherent multipotency of a stem-like CD4 T cell population embedded within the sorted CD4 T cell pool. In conjunction with findings from other diseases, stem-like CD4 T cells provide a unifying framework for understanding CD4 T cell differentiation and plasticity.

Proposed models of CD4 T cell differentiation

The data described above, in combination with the cumulative mechanistic experiments on CD4 T cell differentiation over the years, allow us to propose several possible models that encompass observations across disease states (Figure 2). Each model integrates recent findings and explores how stem-like CD4 T cells may fit within known differentiation trajectories, including their potential to generate different effector populations. We evaluate the strengths and limitations of each model based on current evidence and propose experimental strategies to test and refine them. Although not the focus of our discussion, it is worth noting that in each of these models there is evidence that a pool of terminal effector cells directly arises from naïve cells early in the response independently of a stem-like CD4 T cell population [18,52,53,73]. Although we propose that stem-like CD4 T cells emerge early during the response, the precise timing of their development and the full extent of their differentiation potential remain incompletely understood. Analogous to findings in CD8 T cells, where stem-like cells emerge within the first 5 days of infection [68,71], antigen-specific stem-like CD4 T cells are present within tumor-draining lymph nodes as early as 7 days following tumor inoculation, supporting early establishment of this population [54,55]. Further studies will be necessary to determine the timing of the development, maintenance, and differentiation flexibility of these cells across disease states. In addition, although they have only been described in several chronic conditions, it is possible that stem-like CD4 T cells are present among the currently described effector populations across acute immune responses. We propose that the strength of T cell receptor (TCR) stimulation and initial activating signals in the environment impact the initial rate of differentiation towards effector or stem-like fates. However, our discussion is focused on the established stem-like CD4 T cell population after initial antigen encounter.

Model A – linear differentiation of stem-like CD4 T cells

The first model postulates that naïve CD4 T cells generate a pool of stem-like CD4 T cells which serve as precursors for all effector CD4 T cell helper subsets. In this linear model, a small population of pluripotent lineage negative stem-like CD4 T cells would be expected in every disease

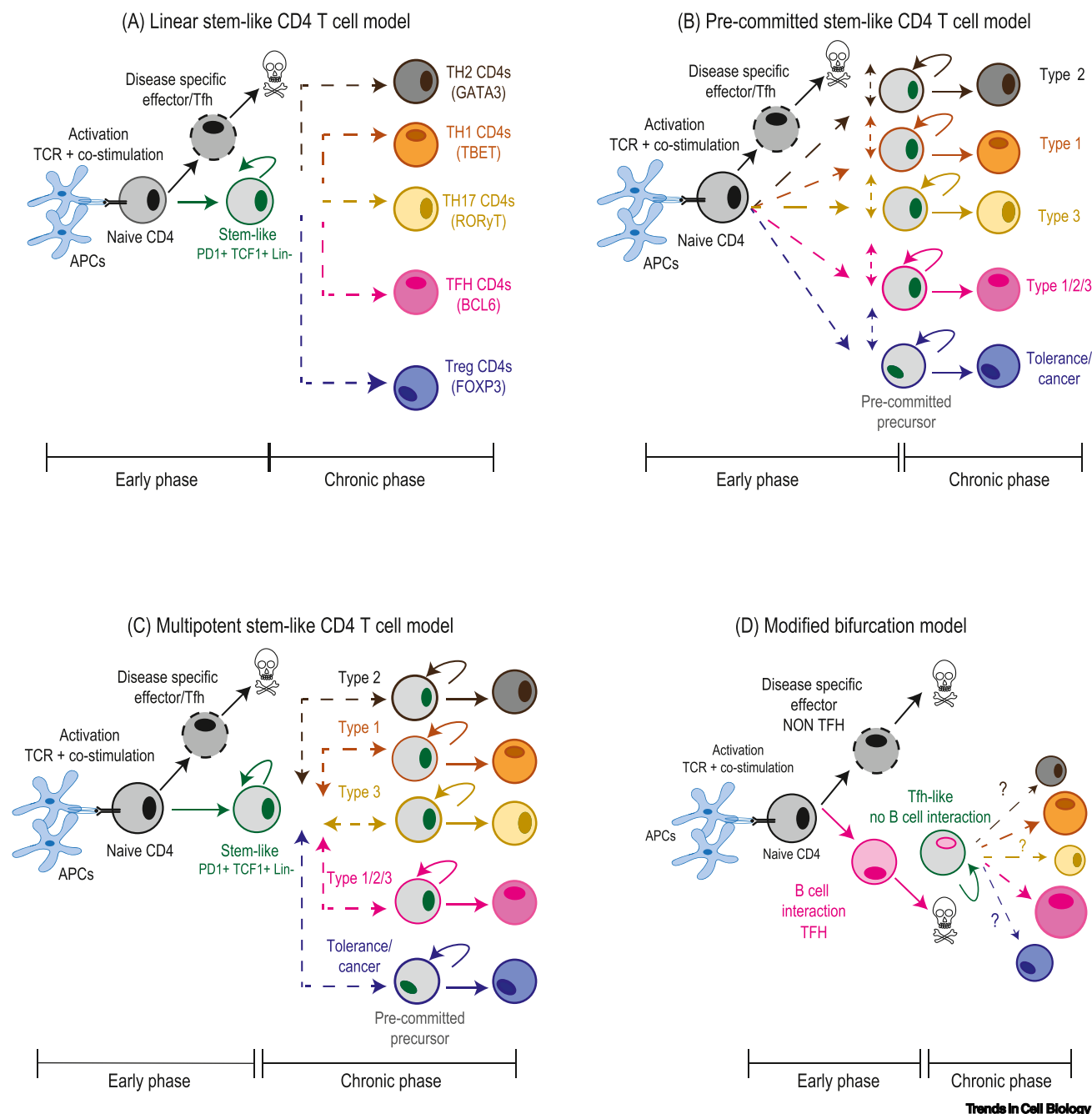


Figure 2. Possible models of chronic CD4 T cell differentiation. (A) In the linear stem-like CD4 T cell model, naïve CD4 T cells activate in the presence of cognate antigen and costimulatory signals, resulting in the generation of disease specific effector/T follicular helper (Tfh) cells and stem-like CD4 T cells. Stem-like CD4 T cells then self-renew and give rise to terminal effector subsets based on cytokine signals in the environment. This population would maintain proliferative and multipotent differentiation capacity to sustain the long-term CD4 T cell response. In this model, stem-like CD4 T cells established during the early phase of infection retain an uncommitted phenotype and transcriptional state through the chronic phase. (B) In the precommitted stem-like CD4 T cell model, naïve CD4 T cells can give rise to disease specific effector/Tfh cells and long-lived precommitted precursor cells with biased differentiation fates. The precommitted precursor CD4 T cells established in the early phase of the response would preferentially give rise to terminal effector CD4 T cells related to that lineage. However, this differentiation model suggests that the precommitted precursor populations retain the potential to directly undergo alternative differentiation if placed in a different environment. (C) The decreasing level of multipotency model suggests that a naïve CD4 T cell generates a heterogeneous pool of terminal effector/Tfh CD4 T cells and a stem-like CD4 T cell population. The

(Figure legend continued at the bottom of the next page.)

setting, giving rise to largely clonally expanded effector CD4 T cells populations across tissues (Figure 2A). In this model, stem-like CD4 T cells retain long-term multipotency and the capacity to adapt and differentiate into alternative CD4 T cell effector fates if exposed to a new environment. This model is supported by evidence from non-inflammatory or low-affinity antigen responses, such as cancer and food tolerance, where antigen-specific CD4 T cells preferentially generate lineage-negative multipotent stem-like CD4 T cells [7,54,74]. These data together with experiments from LCMV, where virus-specific stem-like CD4 T cells gave rise to both Th1 and Tfh effector populations [46,55], suggest that multipotency is conserved across models. This framework could explain the plasticity observed in transfer experiments where the expansion and differentiation of contaminating stem-like CD4 T cells among sorted populations outcompeted terminal effectors, giving the appearance that fully committed CD4 T cells had reprogrammed into alternative lineages. However, a limitation of this model is the ability to reconcile findings from reporter-tracing experiments where cells previously expressing a lineage-defining transcription factor can repress the program and initiate the expression of an alternative master regulator, thereby adopting a new differentiation fate [75–77]. Thus, while some aspects of this model likely contribute to CD4 T cell differentiation, it does not fully capture the complexity of lineage plasticity across different settings. Lineage-tracing and transfer experiments incorporating stem-like and effector CD4 T cells may help to resolve these observations.

Model B – precommitted stem-like CD4 T cells

This model proposes that partially committed stem-like CD4 T cells are generated in response to each disease environment. These precommitted precursor cells would have a strong differentiation bias toward the effector lineage in each disease context while still retaining some degree of multipotency to adopt alternative fates. We would expect these precommitted precursors to exhibit some features of the terminal state without undergoing full differentiation. One feature might be that these precommitted precursor cells express low levels of lineage-defining transcription factors or cytokines while simultaneously maintaining the core stem-like program (TCF7, SLAMF6, ID3). This model predicts that, over time, precommitted precursor cells progressively lose pluripotency as their differentiation bias is further stabilized (Figure 2B). This framework could explain the observed plasticity in some models; if precommitted precursor CD4 T cells were included in transfer experiments, they could have given rise to alternative effector subsets based on new local cytokine signals. In addition, this model provides a potential explanation for hybrid phenotypes with mixed transcriptional lineage features, such as TCF1⁺ FOXP3⁺ cells or RORγT⁺ IFN-γ⁺ cells, which may exist in quasi-stable states [59,60,78,79]. Unlike the linear differentiation model, this framework is consistent with lineage-tracing experiments where precommitted precursors previously expressing a master transcription factor could repress the established program and adopt new alternative fates under specific conditions [35,40,78,79]. However, this model does not explain the presence of lineage-negative stem-like cells observed in several systems [7,54,74]. This model would predict that, upon transfer, a Th1-precommitted precursor cell would give rise to an alternative precommitted precursor cell without the establishment of a lineage-negative stem-like cell. To functionally test this model, precommitted, stem-like,

stem-like population in this model is established early during priming and sustains pluripotency throughout the response. These stem-like CD4 T cells then give rise to precommitted precursor CD4 T cells that preferentially give rise to terminal effector CD4 T cells within that lineage. In contrast to model B, precommitted precursors return to a lineage-negative stem-like fate and reacquire new precommitted properties to differentiate into an alternative fate. (D) The modified bifurcation model proposes that naïve CD4 T cells undergo a bifurcation event upon activation, generating a simultaneous pool of non-Tfh effector cells and Tfh-prone cells based on T cell receptor (TCR) signal strength and interleukin-2 signaling. If the Tfh-prone population migrates to the T–B border and interacts with antigen-specific B cells, it undergoes full Tfh differentiation through intrinsic signaling, ultimately leading to BCL6 expression. If Tfh-prone CD4 T cells do not interact with B cells, however, they maintain a Tfh-like fate with self-renewal and some multipotent differentiation capacity. In contrast to all other models, in this framework all the CD4 T cell plasticity is strictly encompassed within the Tfh-prone arm. Abbreviations: APC, antigen-presenting cell; PD1, programmed death protein 1; TCF1, T cell factor 1; TH1, TH2, TH17, type 1/2/17 T helper cells; Treg, regulatory T cell.

and terminally differentiated cells could be isolated from one condition and transferred into an alternative disease model with the same antigen to challenge their fate commitment and multipotency.

Model C – decreasing levels of multipotency

This model represents a hybrid of models A and B, and proposes that naïve CD4 T cells generate both a small population of stem-like cells and a larger pool of precommitted precursor cells. In this framework, stem-like cells retain full multipotency while precommitted precursor cells are generally restricted to their designated fate. In this model, we would expect precommitted precursors to express low levels of lineage-specific transcription factors, while stem-like cells would remain in an uncommitted lineage negative state. This model accounts for many of the observations addressed by model B, but with one key distinction: it predicts that instead of directly transitioning between partially committed states, precursor cells may revert to a stem-like state before adopting a new differentiation fate (Figure 2C). While this framework does not fully explain the existence of hybrid CD4 T cells that simultaneously express multiple lineage-associated transcription factors, it aligns well with findings from LCMV clone-13 infection, cancer, and tolerance models where stem-like lineage negative CD4 T cells retain substantial multipotency to differentiate into various effector populations. Although the precommitted precursor states have not been defined, evidence for Th1-like and Tfh-like cells that express low levels lineage-specific markers support their existence within this model [34,52,54]. Experimentally, this model suggests that a precommitted precursor cell from one disease model, for example a Th17-biased (type-3) precommitted precursor, would undertake a lineage-negative stem-like state before establishing a new precommitted precursor population in a new environment with a distinct differentiation bias. This stepwise transition through a stem-like population provides a potential mechanism to explain plasticity without requiring direct transitions between committed effector cells. To functionally test this model, lineage-tracing and retransfer experiments of precommitted precursor and stem-like CD4 T cells across different disease settings could be performed.

Model D – modified bifurcation and pro-Tfh cell plasticity

This model builds on previous early bifurcation frameworks of CD4 T cell differentiation [3,51]. In this model, we propose that an early binary decision results in naïve CD4 T cells generating simultaneous pools of disease-specific effector cells and Tfh-prone cells. In contrast to all the other models, the modified bifurcation model proposes that the differentiation plasticity is largely retained within the Tfh-prone arm. Upon interaction with antigen-specific B cells at the T–B border, Tfh-biased CD4 T cells receive the necessary signals to fully commit to the Tfh lineage [18,34,52]. However, some pro-Tfh cells may fail to receive these B cell-dependent signals or localize properly within B cell follicles, potentially due to alternative chemokine receptor expression or tissue localization constraints. As a result, these Tfh-like cells remain in a committed but flexible differentiation state (Figure 2D). Support for this model comes from LCMV infection experiments where the Tfh population gave rise to both Tfh and Th1 subsets, whereas the Th1 population largely retained its phenotype upon transfer [46,55]. However, given the similarities in phenotype between stem-like and differentiated Tfh cells, it is likely that the transferred Tfh population encompasses both pro-Tfh and fully differentiated Tfh cells. Recent findings also suggest that, although BCL6 is not expressed by stem-like CD4 T cells, it may be required for their development [55]. In the context of new research to identify stem-like CD4 T cells in various models, we believe that it is crucial to understand whether the pro-Tfh cells proposed in this bifurcation model are similar cells or whether they exist independently of *bona fide* stem-like CD4 T cells.

Overall, these models provide new frameworks that begin to explain the heterogeneity of antigen-specific CD4 T cell populations, particularly under conditions of persistent stimulation. We

propose that stem-like CD4 T cells provide some context to features of CD4 T cell plasticity and provide an explanation for sustained CD4 T cell effector responses across diseases.

Concluding remarks

The continuous differentiation of stem-like CD4 T cells into helper subsets determines immune outcomes across infections, cancer, transplantation, and autoimmunity. Therefore, harnessing their remarkable capacity to retain differentiation plasticity represents a promising therapeutic avenue.

As proof of this concept, redirecting tumor specific stem-like CD4 T cells from a restricted Treg fate toward an active Th1 program was sufficient to rescue effective CD8 T cell antitumor immunity and enhance response to checkpoint blockade therapy within the suppressive cancer environment. This effect was not confined to secondary lymphoid tissues given that Th1 cells generated from these stem-like CD4 T cells infiltrated the tumor and promoted an effective CD8 T cell effector response, ultimately leading to tumor control. Notably, once stem-like CD4 T cells transitioned from a 'restricted' Treg fate to an 'active' Th1 program, this new trajectory appeared to be self-reinforced by the endogenous stem-like CD4 T cell pool [54]. These findings suggest that their differentiation fate is not locked but instead is responsive and adaptable to the environmental cues, underscoring the therapeutic potential of harnessing the differentiation flexibility of stem-like CD4 T cells that can have significant impacts on the outcome of the immune response.

Moving forward, a major challenge lies in identifying the signals that allow stem-like CD4 T cells to retain multipotency despite chronic stimulation, as well as those that can drive them into alternative fates across disease settings (see [Outstanding questions](#)). These findings could be used to design adoptive cell transfer therapies with enhanced persistence and adaptability. Moreover, clarifying whether distinct stem-like CD4 T cell populations emerge under different environments will be crucial to target these populations *in vivo*. The ability to redirect an antigen specific population toward proinflammatory or suppressive fates opens new opportunities for immunotherapies. For example, promoting a Th1 fate in cancer could synergize with current checkpoint blockade therapies to enhance CD8 T cell responses. Conversely, redirecting stem-like CD4 T cells toward Treg fates in transplantation may suppress immune-mediated pathology and donor-specific antibody formation, thereby preventing organ rejection. Stem-like CD4 T cells therefore represent an exciting and novel target for immune regulation to either enhance protective immunity or repress immune pathology. Defining the signals and transcriptional frameworks that underlie stem-like CD4 T cell differentiation will be crucial to leverage their therapeutic potential across disease settings.

Acknowledgments

This work was supported by the James M. Cox Foundation, funding from the Prostate Cancer Foundation, pilot funding from the Winship Cancer Institute supported by the Dunwoody Country Club Senior Men's Association, the National Cancer Institute (NCI grant 1R01CA280069) and the Cancer Research Institute Lloyd J. Old STAR program.

Declaration of interests

The authors declare no competing interests.

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Outstanding questions

When are stem-like CD4 T cells established during the response?

What factors control the development and maintenance of the stem-like CD4 T cell fate across disease settings?

What mechanisms regulate stem-like CD4 T cell fate towards helper or regulatory fates?

Are stem-like CD4 T cells precommitted, or do they sustain multipotency in each disease setting?

Does the stem-like CD4 T cell transcriptional and epigenetic landscape change over time within each condition?

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